

EXECUTIVE SUMMARY

Customer Churn Prediction & Revenue Protection Strategy

Business Problem

Customer churn, currently at **10.60%**, is reducing recurring revenue and increasing acquisition costs due to the absence of a proactive risk identification system. This project identified key churn drivers, built a predictive model to detect high-risk customers, and estimated revenue exposure to support targeted, data-driven retention strategies.

Key Findings

Data analysis revealed statistically significant churn patterns (T-test $p < 0.001$; Chi-square $p < 0.001$):

- Customers with short tenure are significantly more likely to churn.
- Month-to-month contracts have the highest churn rates.
- Customers with higher monthly charges demonstrate increased churn probability.
- Certain payment methods are associated with elevated churn behavior.

A tuned Random Forest model (max_depth=5, n_estimators=100) achieved **0.99 ROC-AUC** and **95% accuracy**, effectively identifying high-risk customers for proactive retention.

Financial Impact

Revenue analysis indicates that high-risk customers represent measurable annual exposure, with **estimated revenue at risk of 10,212**. Even a 5% reduction in churn would protect meaningful recurring revenue, enhance customer lifetime value, and generate strong return on investment through targeted retention initiatives.

Recommendations

1. **Contract Optimization:** Shift month-to-month customers to long-term plans.
2. **Early Engagement:** Strengthen onboarding for customers within their first 12 months.
3. **Pricing Strategy:** Offer selective incentives to high-risk segments.
4. **Risk Automation:** Integrate churn scoring into CRM for proactive retention actions

Implementation Plan

Phase	Deliverable / Outcome
Day 1 - 2	Model deployed; KPI dashboard operational
Day 3 - 4	Targeted retention campaigns launched
Day 5 - 6	Churn impact and revenue protection evaluated
Day 7	Model optimized and performance validated